

AI for Science at JHU/APL

Access, Execution and Verification at the Jagged Frontier

Speaker: Mike Wolmetz, PhD

Source: JHUAPL_AIforScience_092126.pdf, pasted-text.txt

Core thesis: AI is advancing science unevenly; practical impact depends on access, execution, and scalable verification.

AI Index

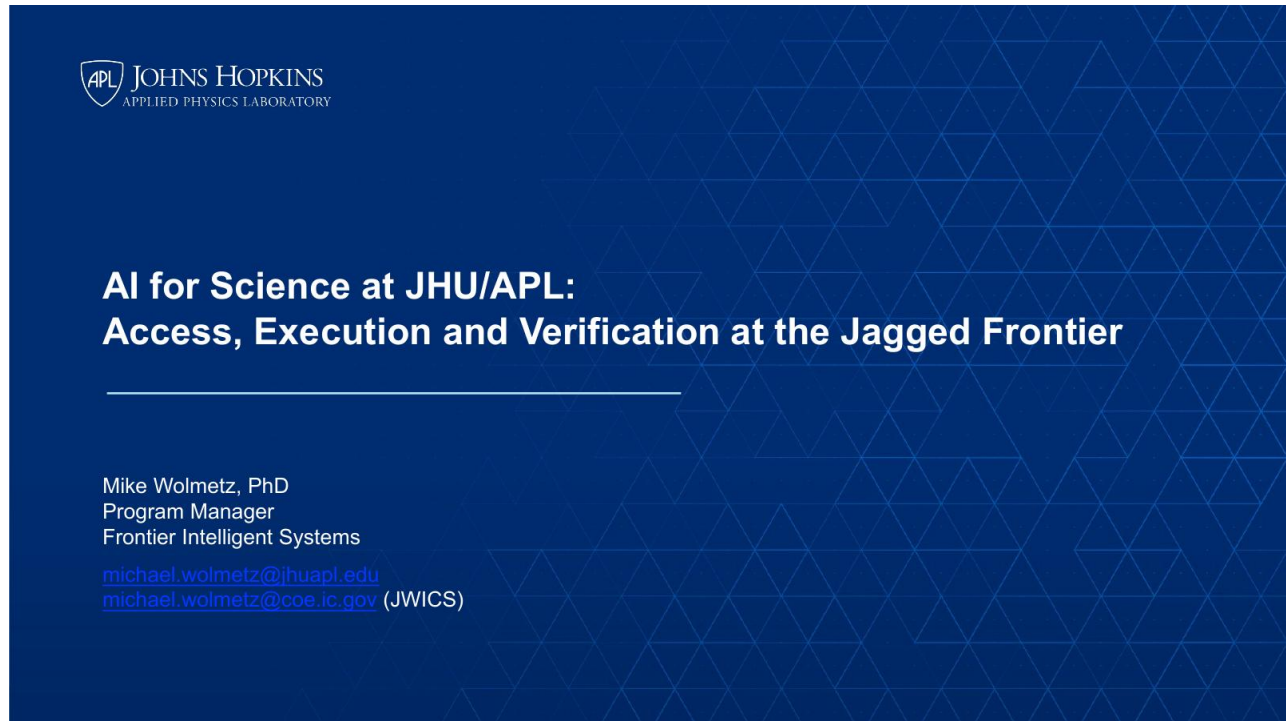
Key topics: AI for science; tacit knowledge; autonomous laboratories; materials discovery; verification; replication

Major outcomes: LLM-assisted novice protocol: 13.5/18 versus 4/18; 485 autonomous electrolyte formulations; More than 2x benchmark capacity at -40 C; 19 of 200 feasibility judgments revised

Open questions: Frontier-model improvement on tacit laboratory knowledge; Automation of final testing and complex characterization; Scaling replication beyond digital experiments

Note: Claims, figures, and links are represented from the supplied deck and transcript; they were not independently verified.

01-introduction-and-jagged-frontier | Introduction and jagged frontier



Slide content: Title and speaker information.

What was said: AI-for-science progress is present but unevenly distributed.

Key quote/result/takeaway: AI is advancing science unevenly.

Entities: Mike Wolmetz; Frontier Intelligent Systems; APL

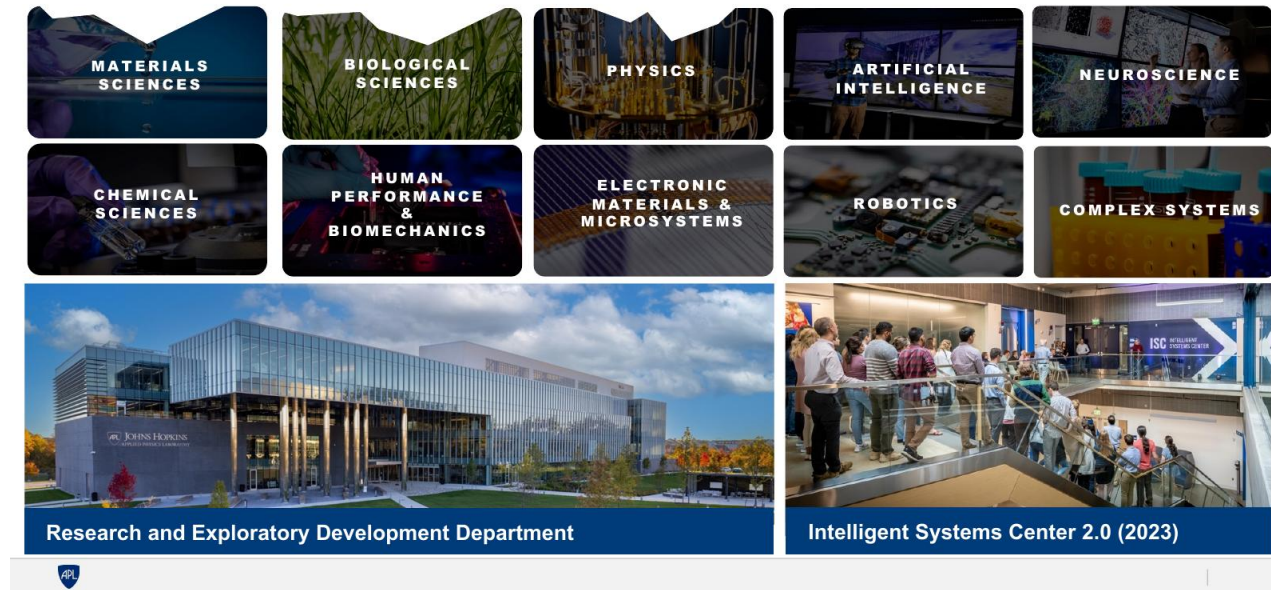
Claims: Frontier Intelligent Systems spans AI, robotics, complex systems, and neuroscience.

Evidence/results: Speaker introduction and title slide.

Source confidence: high: Direct speaker statement.

02-apl-context-and-cross-domain-work | APL context and cross-domain work

Science is Changing Rapidly but Jaggedly



Slide content: APL technical domains and Intelligent Systems Center 2.0.

What was said: Co-location exposes uneven AI transformation across domains.

Key quote/result/takeaway: The transformation is fast but jagged.

Entities: APL; Intelligent Systems Center

Claims: Cross-domain proximity enables incidental collaboration.

Evidence/results: Domain map and organizational description.

Source confidence: high: Directly presented context.

03-access-execution-verification-framework | Access, execution, and verification

Why the AI for Science Frontier is Jagged

	 1 Access	 2 Execution	 3 Verification
AI Enabler	Interact with relevant literature, data, prior results, models, tools, and context	Run simulations or safely invoke laboratory actions through a controlled interface	Produce results with evidence that can be inspected, interpreted, and verified at scale
AI Limiter	<i>Most laboratory science involves significant tacit knowledge that may not be accessible</i>	<i>Some domains are more amenable to simulation and lab automation than others</i>	<i>Efficient verification via physical measurement, simulation, or expert assessment is a bottleneck</i>
			

Slide content: Three enablers and their limiters.

What was said: Tacit knowledge limits access, automation varies execution, and verification is likely to bottleneck.

Key quote/result/takeaway: Verification will increasingly bottleneck agentic science.

Entities: intelligent systems; laboratory science

Claims: Tacit knowledge and scalable verification constrain AI-for-science.

Evidence/results: Framework slide.

Source confidence: high: Central framing of talk.

04-access-tacit-knowledge-in-biology | Tacit knowledge in biology

1 Access

TASK: Prepare a *Bacillus thuringiensis* spore preparation and quantify its purity. Describe what the results will look like.

NON-EXPERT	NON-EXPERT + LLM
4 / 18	13.5 / 18
"NON-VIABLE"	"Extremely short protocol that would require an extreme amount of tacit knowledge to complete successfully."

The image displays a laboratory setting with a scientist in a white lab coat, blue hairnet, and safety glasses. Overlaid on this is a task description and two columns of expert scores. The task is to prepare a *Bacillus thuringiensis* spore preparation and quantify its purity. The left column, labeled 'NON-EXPERT', shows a score of 4/18 and a red callout stating 'NON-VIABLE'. The right column, labeled 'NON-EXPERT + LLM', shows a score of 13.5/18 and a blue callout stating 'Extremely short protocol that would require an extreme amount of tacit knowledge to complete successfully.' The background image is a composite of a scientist working and a molecular structure.

Slide content: Bacillus protocol task and expert scores.

What was said: LLMs improved novice protocol scores but omitted practical laboratory knowledge.

Key quote/result/takeaway: Models omit the tacit knowledge needed for execution.

Entities: DARPA; Bacillus thuringiensis; LLMs

Claims: LLM assistance raised a novice protocol score from 4/18 to 13.5/18.

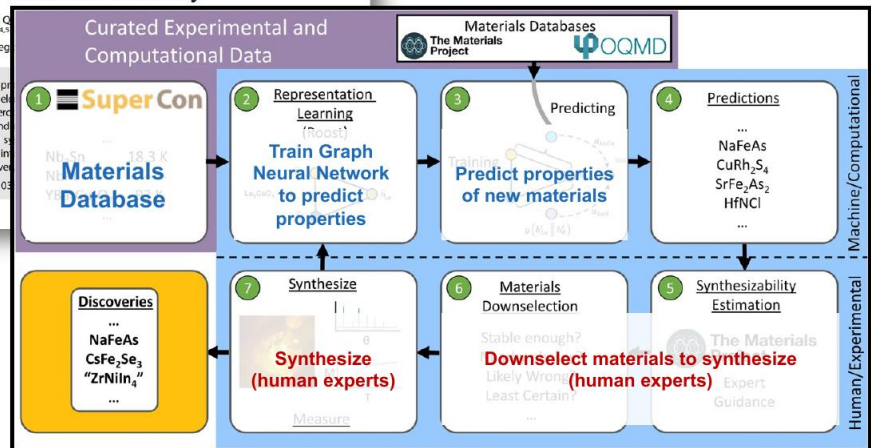
Evidence/results: Expert scores and evaluator quote.

Source confidence: high: Reported study result.

05-execution-2023-superconductor-loop | 2023 superconductor loop

Starting to Close the Loop in Materials

SOTA in 2023
Superconductor discovery



Pogue, L., et al (2023). Closed-loop superconducting materials discovery. npj Computational Materials 9:181

Slide content: GNN screening paired with human downselection and synthesis.

What was said: ML accelerated search but people manually closed the experimental loop.

Key quote/result/takeaway: Humans closed the loop by hand.

Entities: SuperCon; graph neural network

Claims: AI screened hundreds of thousands of compositions.

Evidence/results: Reported doubled hit rate and novel superconductor.

Source confidence: medium: Reported result; study not independently reviewed.

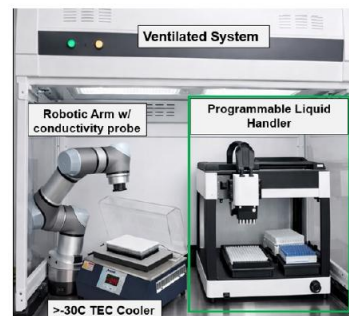
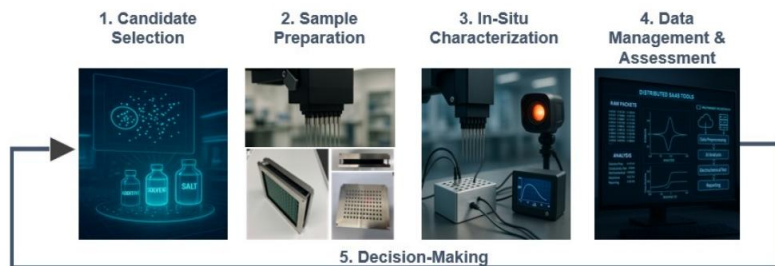
06-execution-autonomous-battery-discovery | Autonomous battery discovery

Autonomous Closed-Loop Synthesis and Characterization

SOTA Today
Battery Electrolyte Discovery



“Generate a hypothesis and test plan to improve low temperature performance of lithium ion batteries through new electrolyte design”



Outcomes

- Designed and autonomously screened 485 battery electrolyte formulations
- System reduces researcher effort from ~44 hours of manual experimentation to ~1.5 hours experiment design and system setup.
- Top candidate delivers over twice the capacity of leading benchmarks at -40 °C

AI Limiter

- Final testing done manually by constructing cells with each electrolyte
- Requires dexterity and judgment beyond today's robotic autonomy



Slide content: Autonomous battery-electrolyte formulation and results.

What was said: Automation cut effort but not final manual cell construction.

Key quote/result/takeaway: 485 formulations and more than 2x benchmark capacity at -40 C.

Entities: lithium-ion batteries; lab robotics

Claims: Autonomous screening reduced reported effort from about 44 to 1.5 hours.

Evidence/results: Slide-reported count, capacity, and time figures.

Source confidence: high: Directly reported on slide.

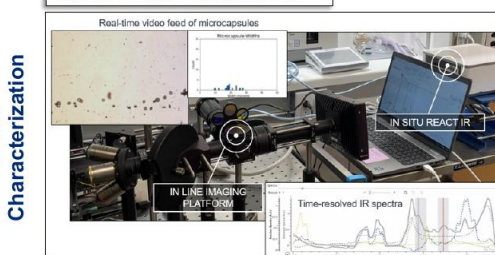
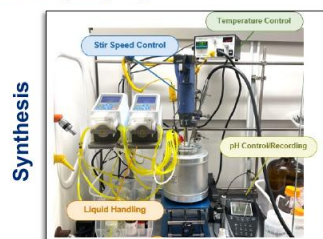
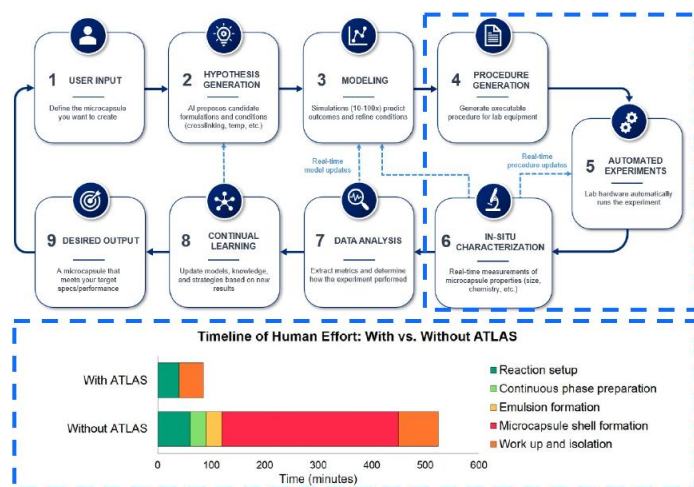
07-execution-microcapsule-discovery | Autonomous microcapsule discovery

Autonomous Closed-Loop Synthesis and Characterization

SOTA Today
Microcapsule Discovery



"Make a microcapsule that encapsulates mineral oil and is robust to spraying"



<https://www.jhuapl.edu/news/news-releases/260309-microcapsule-atlas>

Slide content: ATLAS closed loop for robust mineral-oil microcapsules.

What was said: Automation sped synthesis and characterization, but complex in-situ characterization remains difficult.

Key quote/result/takeaway: More results are arriving on the first try.

Entities: ATLAS; microcapsules

Claims: The process was reported as about an order of magnitude faster.

Evidence/results: Workflow and timeline comparison.

Source confidence: medium: Detailed measures not provided.

08-verification-scientific-feasibility | Scientific feasibility verification

Human and AI SMEs



1 Access



3 Verification

Sample materials science claim: "An aluminum alloy with thermally stable nanometer-scale dispersoids can maintain microstructures with grain sizes of 10 microns or less during superplastic forming, enabling elongation beyond 300% without premature loss of ductility."



[SFBench: The SciFy Scientific Feasibility Benchmark](#)



[DARPA built an AI to fact-check enemy weapons claims | SciAm](#)



12

Slide content: SFBench and DARPA SciFy materials-claim verification.

What was said: Systems changed expert ground truth on 19 of 200 claims.

Key quote/result/takeaway: Verification is hard for humans and AI.

Entities: DARPA SciFy; SFBench; Erica Briscoe

Claims: AI feedback prompted expert reassessment of feasibility claims.

Evidence/results: 19 of 200 claims reportedly revised.

Source confidence: high: Direct speaker report.

09-automating-replication-and-research | Automating replication

Automating Replication



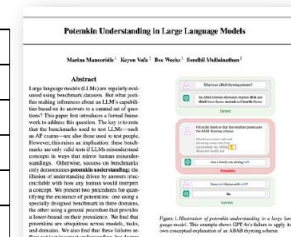
AutoReplication Cookbook v0.1

1. Ingest material: load the paper, code, and data
2. Select Claim: pick the specific result to test
3. Plan: map out how to test it
4. Implement: write or adapt the code
5. Test/validate: run it and compare to the paper
6. Aggregate & Report: summarize matches, gaps, and assumptions



Topic	Last commit	Last update
1. ingest-material	Updated to test and compare alignments...	10/24/24
2. select-claim	Updated to test and compare alignments...	10/24/24
3. plan	Updated to test and compare alignments...	10/24/24
4. implement	Updated to test and compare alignments...	10/24/24
5. test/validate	Updated to test and compare alignments...	10/24/24
6. aggregate-report	Updated to test and compare alignments...	10/24/24

Category	Papers
AI/ML, agents, and robotics	15
Statistics/economics/social science	5
Physical science/materials	3
Numerical/scientific computing	1
Classical computer vision	1



13

Slide content: Six-step AutoReplication Cookbook and hackathon portfolio.

What was said: A failed digital replication produced a new hypothesis for agentic investigation.

Key quote/result/takeaway: Auto-replication leads to auto-research.

Entities: AutoReplication Cookbook; Codex; Potemkin Understanding

Claims: The effort focused on digitally reproducible papers.

Evidence/results: 25-paper category table and hackathon account.

Source confidence: high: Direct speaker report.

10-capability-overhang-and-mission-context | Capability overhang and mission context



Slide content: APL facilities across scientific and engineering domains.

What was said: Access, execution, and verification are needed to realize present capability; mission intent shapes UARC research early.

Key quote/result/takeaway: Capability exceeds organizational ability to use it.

Entities: APL; UARC; Department of War; intelligence community

Claims: Mission intent and bespoke constraints influence early research.

Evidence/results: Closing and Q&A remarks.

Source confidence: high: Speaker perspective.

High-level Takeaways

- Use access, execution, and verification to assess AI-for-science maturity.
- Tacit laboratory knowledge remains a major access barrier.
- Autonomous labs accelerate parts of discovery while leaving physical execution gaps.
- Verification is a central scaling constraint.
- Replication can become a generator of new agentic research.

Processing Notes

- Slide images were rendered locally from the supplied 15-page PDF.
- Claims and figures were preserved from supplied sources and not independently verified.
- Topical alignment follows slide order because no timestamps were supplied.